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# Vivado v2025.2 (64-bit)
# SW Build 6299465 on Fri Nov 14 19:35:11 GMT 2025
# IP Build 6300035 on Fri Nov 14 10:48:45 MST 2025
# SharedData Build 6298862 on Thu Nov 13 04:50:51 MST 2025
# Start of session at: Sat Jul 18 22:56:24 2026
# Process ID           : 33200
# Current directory    : D:/Vivado/MMA/MMA.runs/synth_1
# Command line         : vivado.exe -log mat_accel.vds -product
Vivado -mode batch -messageDb vivado.pb -notrace -source
mat_accel.tcl
# Log file             : D:/Vivado/MMA/MMA.runs/synth_1/mat_accel.vds
# Journal file         : D:/Vivado/MMA/MMA.runs/synth_1\vivado.jou
# Running On           : AK_LAPTOP
# Platform             : Windows Server 2016 or Windows 10
# Operating System     : 26200
# Processor Detail     : Intel(R) Core(TM) i5-10300H CPU @ 2.50GHz
# CPU Frequency        : 2496 MHz
# CPU Physical cores   : 4
# CPU Logical cores    : 8
# Host memory          : 8419 MB
# Swap memory          : 13958 MB
# Total Virtual        : 22377 MB
# Available Virtual    : 3347 MB
#-----
source mat_accel.tcl -notrace
create_project: Time (s): cpu = 00:00:10 ; elapsed = 00:00:08 .
Memory (MB): peak = 525.871 ; gain = 220.012
Command: synth_design -top mat_accel -part xc7a12tcs325-1
Starting synth_design
Attempting to get a license for feature 'Synthesis' and/or device
'xc7a12t'
INFO: [Common 17-349] Got license for feature 'Synthesis' and/or
device 'xc7a12t'
INFO: [Device 21-403] Loading part xc7a12tcs325-1
INFO: [Synth 8-7079] Multithreading enabled for synth_design using
a maximum of 2 processes.
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INFO: [Synth 8-7078] Launching helper process for spawning children
vivado processes
INFO: [Synth 8-7075] Helper process launched with PID 6204
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Starting RTL Elaboration : Time (s): cpu = 00:00:06 ; elapsed =
00:00:07 . Memory (MB): peak = 1242.727 ; gain = 540.148
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INFO: [Synth 8-6157] synthesizing module 'mat_accel'
[D:/Projects/MMA/codes/mat_accel.v:1]
INFO: [Synth 8-6157] synthesizing module 'control_fsm'
[D:/Projects/MMA/codes/control_fsm.v:1]
INFO: [Synth 8-155] case statement is not full and has no default
[D:/Projects/MMA/codes/control_fsm.v:82]
INFO: [Synth 8-6155] done synthesizing module 'control_fsm' (0#1)
[D:/Projects/MMA/codes/control_fsm.v:1]
INFO: [Synth 8-6157] synthesizing module 'pe'
[D:/Projects/MMA/codes/pe.v:1]
    Parameter WIDTH bound to: 8 - type: integer
INFO: [Synth 8-6157] synthesizing module 'mac'
[D:/Projects/MMA/codes/mac.v:1]
    Parameter WIDTH bound to: 8 - type: integer
INFO: [Synth 8-6157] synthesizing module 'multiplier'
[D:/Projects/MMA/codes/multiplier.v:1]
    Parameter WIDTH bound to: 8 - type: integer
INFO: [Synth 8-6155] done synthesizing module 'multiplier' (0#1)
[D:/Projects/MMA/codes/multiplier.v:1]
INFO: [Synth 8-6155] done synthesizing module 'mac' (0#1)
[D:/Projects/MMA/codes/mac.v:1]
INFO: [Synth 8-6155] done synthesizing module 'pe' (0#1)
[D:/Projects/MMA/codes/pe.v:1]
INFO: [Synth 8-6155] done synthesizing module 'mat_accel' (0#1)
[D:/Projects/MMA/codes/mat_accel.v:1]
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Finished RTL Elaboration : Time (s): cpu = 00:00:08 ; elapsed =
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00:00:09 . Memory (MB): peak = 1360.027 ; gain = 657.449

Start Handling Custom Attributes

Finished Handling Custom Attributes : Time (s): cpu = 00:00:08 ;
elapsed = 00:00:09 . Memory (MB): peak = 1360.027 ; gain = 657.449

Finished RTL Optimization Phase 1 : Time (s): cpu = 00:00:08 ;
elapsed = 00:00:09 . Memory (MB): peak = 1360.027 ; gain = 657.449

Netlist sorting complete. Time (s): cpu = 00:00:00 ; elapsed =
00:00:00.006 . Memory (MB): peak = 1360.027 ; gain = 0.000
INFO: [Project 1-570] Preparing netlist for logic optimization

Processing XDC Constraints

Initializing timing engine

Parsing XDC File

[D:/Vivado/MMA/MMA.srscs/constrs_1/new/clock_constraint.xdc]

Finished Parsing XDC File

[D:/Vivado/MMA/MMA.srscs/constrs_1/new/clock_constraint.xdc]

Completed Processing XDC Constraints

Netlist sorting complete. Time (s): cpu = 00:00:00 ; elapsed =
00:00:00 . Memory (MB): peak = 1429.258 ; gain = 0.000

INFO: [Project 1-111] Unisim Transformation Summary:

No Unisim elements were transformed.

Constraint Validation Runtime : Time (s): cpu = 00:00:00 ; elapsed
= 00:00:00.021 . Memory (MB): peak = 1429.258 ; gain = 0.000

Finished Constraint Validation : Time (s): cpu = 00:00:17 ; elapsed
= 00:00:19 . Memory (MB): peak = 1429.258 ; gain = 726.680

Start Loading Part and Timing Information

Loading part: xc7a12tcsg325-1

Finished Loading Part and Timing Information : Time (s): cpu =
00:00:17 ; elapsed = 00:00:19 . Memory (MB): peak = 1429.258 ; gain
= 726.680

Start Applying 'set_property' XDC Constraints

Finished applying 'set_property' XDC Constraints : Time (s): cpu =
00:00:17 ; elapsed = 00:00:19 . Memory (MB): peak = 1429.258 ; gain
= 726.680

INFO: [Synth 8-802] inferred FSM for state register 'state_reg' in
module 'control_fsm'

State	New Encoding
Previous Encoding	

IDLE	00001
000	
CLEAR	00010
001	
CYCLE1	00100
010	
CYCLE2	01000
011	
DONE	10000
100	

INFO: [Synth 8-3354] encoded FSM with state register 'state_reg' using encoding 'one-hot' in module 'control_fsm'

Finished RTL Optimization Phase 2 : Time (s): cpu = 00:00:17 ; elapsed = 00:00:19 . Memory (MB): peak = 1429.258 ; gain = 726.680

Start RTL Component Statistics

Detailed RTL Component Info :

+---Muxes :

2 Input	8 Bit	Muxes := 4
5 Input	5 Bit	Muxes := 1
2 Input	5 Bit	Muxes := 1
5 Input	1 Bit	Muxes := 1
2 Input	1 Bit	Muxes := 4

Finished RTL Component Statistics

Start Part Resource Summary

Part Resources:

DSPs: 40 (col length:40)

BRAMs: 40 (col length: RAMB18 40 RAMB36 20)

Finished Part Resource Summary

Start Cross Boundary and Area Optimization

Finished Cross Boundary and Area Optimization : Time (s): cpu =
00:00:21 ; elapsed = 00:00:24 . Memory (MB): peak = 1436.500 ; gain
= 733.922

Start Applying XDC Timing Constraints

Finished Applying XDC Timing Constraints : Time (s): cpu = 00:00:27

; elapsed = 00:00:30 . Memory (MB): peak = 1563.992 ; gain = 861.414

Start Timing Optimization

Finished Timing Optimization : Time (s): cpu = 00:00:30 ; elapsed = 00:00:32 . Memory (MB): peak = 1638.684 ; gain = 936.105

Start Technology Mapping

Finished Technology Mapping : Time (s): cpu = 00:00:30 ; elapsed = 00:00:33 . Memory (MB): peak = 1638.738 ; gain = 936.160

Start IO Insertion

Start Flattening Before IO Insertion

Finished Flattening Before IO Insertion

Start Final Netlist Cleanup

Finished Final Netlist Cleanup

Finished IO Insertion : Time (s): cpu = 00:00:35 ; elapsed =
00:00:38 . Memory (MB): peak = 1836.895 ; gain = 1134.316

Start Renaming Generated Instances

Finished Renaming Generated Instances : Time (s): cpu = 00:00:35 ;
elapsed = 00:00:38 . Memory (MB): peak = 1836.895 ; gain = 1134.316

Start Rebuilding User Hierarchy

Finished Rebuilding User Hierarchy : Time (s): cpu = 00:00:35 ;
elapsed = 00:00:38 . Memory (MB): peak = 1836.895 ; gain = 1134.316

Start Renaming Generated Ports

Finished Renaming Generated Ports : Time (s): cpu = 00:00:35 ;
elapsed = 00:00:38 . Memory (MB): peak = 1836.895 ; gain = 1134.316

Start Handling Custom Attributes

Finished Handling Custom Attributes : Time (s): cpu = 00:00:35 ;
elapsed = 00:00:38 . Memory (MB): peak = 1836.895 ; gain = 1134.316

Start Renaming Generated Nets

Finished Renaming Generated Nets : Time (s): cpu = 00:00:35 ;
elapsed = 00:00:38 . Memory (MB): peak = 1836.895 ; gain = 1134.316

Start Writing Synthesis Report

Report BlackBoxes:

	BlackBox name	Instances

Report Cell Usage:

	Cell	Count
1	BUFG	1
2	CARRY4	60
3	LUT2	83
4	LUT3	33
5	LUT5	96
6	LUT6	248
7	FDRE	73
8	FDSE	1
9	IBUF	67
10	OBUF	69

Finished Writing Synthesis Report : Time (s): cpu = 00:00:35 ;
elapsed = 00:00:38 . Memory (MB): peak = 1836.895 ; gain = 1134.316

Synthesis finished with 0 errors, 0 critical warnings and 0
warnings.

Synthesis Optimization Runtime : Time (s): cpu = 00:00:25 ; elapsed

= 00:00:36 . Memory (MB): peak = 1836.895 ; gain = 1065.086
Synthesis Optimization Complete : Time (s): cpu = 00:00:35 ;
elapsed = 00:00:38 . Memory (MB): peak = 1836.895 ; gain = 1134.316
INFO: [Project 1-571] Translating synthesized netlist
Netlist sorting complete. Time (s): cpu = 00:00:00 ; elapsed =
00:00:00.020 . Memory (MB): peak = 1846.055 ; gain = 0.000
INFO: [Netlist 29-17] Analyzing 60 Unisim elements for replacement
INFO: [Netlist 29-28] Unisim Transformation completed in 0 CPU
seconds
INFO: [Project 1-570] Preparing netlist for logic optimization
INFO: [Opt 31-138] Pushed 0 inverter(s) to 0 load pin(s).
Netlist sorting complete. Time (s): cpu = 00:00:00 ; elapsed =
00:00:00.001 . Memory (MB): peak = 1849.789 ; gain = 0.000
INFO: [Project 1-111] Unisim Transformation Summary:
No Unisim elements were transformed.

Synth Design complete | Checksum: 56754d0a
INFO: [Common 17-83] Releasing license: Synthesis
27 Infos, 0 Warnings, 0 Critical Warnings and 0 Errors encountered.
synth_design completed successfully
synth_design: Time (s): cpu = 00:00:40 ; elapsed = 00:00:45 .
Memory (MB): peak = 1849.789 ; gain = 1323.918
Write ShapeDB Complete: Time (s): cpu = 00:00:00 ; elapsed =
00:00:00.008 . Memory (MB): peak = 1849.789 ; gain = 0.000
INFO: [Common 17-1381] The checkpoint
'D:/Vivado/MMA/MMA.runs/synth_1/mat_accel.dcp' has been generated.
INFO: [Vivado 12-24828] Executing command : report_utilization
-file mat_accel_utilization_synth.rpt -pb
mat_accel_utilization_synth.pb
INFO: [Common 17-206] Exiting Vivado at Sat Jul 18 22:57:23 2026...